Mixed Boundary Data: Two Closures, One Evidence Chain

Project 2 — Direction 2 Five-point FDM for Poisson

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MTH321 Numerical Analysis of ODEs and PDEs

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Roadmap and Takeaways

Question (the consultancy brief): solve a mixed-boundary Poisson problem reliably and justify the numbers.

1. **Model.** Dirichlet on three sides, Neumann *or* Robin on the right edge; manufactured solution $u^* = e^{x+y}$.
2. **Discretisation.** Five-point stencil; two second-order closures for the normal derivative — one-sided difference vs. ghost-point elimination.
3. **Verification.** Row-by-row Taylor defects, refinement tables, falsification attempts that *can* kill a wrong code.
4. **Experiments.** Convergence, matrix structure, solver cost, pure-Neumann null space.

One-line answer: the closure is part of the method — ghost elimination reaches 10^{-4} boundary accuracy with $\sim 10^3$ unknowns; the one-sided closure needs $\sim 10^7$.

Background: where mixed BCs come from

Three physical interfaces, one mathematics:

- **Dirichlet:** wall held at known temperature / conductor at known potential.
- **Neumann:** insulated face or prescribed flux $\partial_n u = q$.
- **Robin:** Newton cooling $-\alpha \partial_n u = h_c(u - u_\infty) \;\Rightarrow\; \partial_n u + \beta u = r$.

Numerical moral: the boundary closure is part of the method. A first-order one-sided row and a second-order ghost row solve *different discrete problems* — visible in the convergence table, the matrix, and the cost of reaching a target accuracy.

Model and manufactured data

PDE: $-\Delta u = f$ on $\Omega = (0,1)^2$.
Boundary data (Direction 2):

- $u = g$ on $x = 0$, $y = 0$, $y = 1$ (including all four corners);
- $\partial_x u = q(y)$ *or* $\partial_x u + \beta u = r(y)$ on $x = 1$;
- depth variant: $\partial_n u = q_N$ on the whole boundary.

Manufactured solution (MMS): $u^* = e^{x+y}$, then

$$f = -2e^{x+y}, \qquad q(y) = e^{1+y}, \qquad r(y) = 3e^{1+y}\ (\beta = 2).$$

Exact solution known $\Rightarrow$ true error, not just residual.

Discretisation: five-point stencil

Interior rows, $h = 1/N$:

$$\frac{4u_{i,j} - u_{i\pm1,j} - u_{i,j\pm1}}{h^2} = f_{i,j} + O(h^2).$$

Dirichlet rows: eliminate known boundary values into the right-hand side.

The interesting row is the right edge $x = 1$. Two closures:

(a) One-sided difference

$$\frac{3u_{i,j} - 4u_{i-1,j} + u_{i-2,j}}{2h} = q_j$$

Local defect $O(h) \Rightarrow$ global order 1.

(b) Ghost-point elimination

$$\frac{u_{i+1,j} - u_{i-1,j}}{2h} = q_j \;\Rightarrow\; u_{i+1,j} = u_{i-1,j} + 2h\, q_j$$

substituted into the PDE row. Local defect $O(h)$ in a thin layer, global order 2.

Robin closure (same architecture)

Ghost relation for Robin $\partial_x u + \beta u = r$ on $x = 1$:

$$\frac{u_{i+1,j} - u_{i-1,j}}{2h} + \beta u_{i,j} = r_j \;\Rightarrow\; u_{i+1,j} = u_{i-1,j} + 2h(r_j - \beta u_{i,j}).$$

Substituted into the PDE row gives the diagonal shift $4/h^2 + 2\beta/h$ and RHS load $+2r_j/h$ — **both rows pass hand-computed row defects** (Section 4 of the report).

Takeaway: nothing about the interior stencil changes; only the closure row does. That is exactly what makes the comparison honest.

Verification 1: row defects (Taylor audit)

Before any convergence table: substitute the exact solution into each assembled row and compare with the Taylor-predicted leading defect.

Row	Predicted leading term	Fitted order	defect/h^p vs. predicted
Interior five-point	$-\frac{h^2}{12}(u^*_{xxxx} + u^*_{yyyy})$	2.000	0.453 vs. 0.453
One-sided BC row	$-\frac{h}{2}u^*_{xx}$	0.995	2.229 vs. 2.241
Ghost PDE row	$+\frac{h}{3}u^*_{xxx}$	0.992	1.482 vs. 1.494
Centred BC condition	$+\frac{h^2}{6}u^*_{xxx}$	2.000	0.747 vs. 0.747

Takeaway: measured constants match the hand Taylor analysis to 3 digits — the assembled rows are exactly the equations we derived.

Verification 2: refinement tables

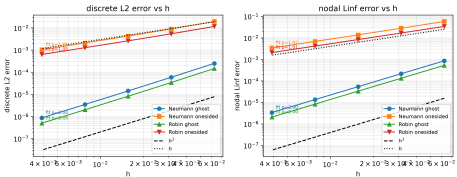
Ghost + Neumann (exact fits):

- L^2 order ≈ 2.0
- L^∞ order ≈ 2.0
- boundary L^∞ order ≈ 2.0

One-sided + Neumann:

- L^2 order ≈ 1.0
- L^∞ order ≈ 1.0

Robin variants behave identically.



Takeaway: orders match pre-registered expectations — and the pairwise rates drift to the fitted ones only on fine grids (documented, not hidden).

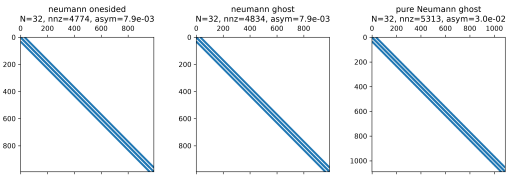
Verification 3: falsification — can we catch a wrong code?

Three deliberate defects injected into the assembly (exp_07):

Run	fitted $p_{h\ell^2}$	fitted p_∞	$\|e\|_\infty$ at N=128
Baseline (correct ghost)	2.052	1.997	1.35×10^{-5}
F1: sign of q flipped	0.038	−0.002	3.48
F2: west coupling not doubled	−0.076	−0.175	6.32
F3: top-edge Dirichlet move dropped	0.040	0.000	7.39

Takeaway: the refinement study is not decoration — each defect destroys the order table in a distinct way. Our tables would have failed.

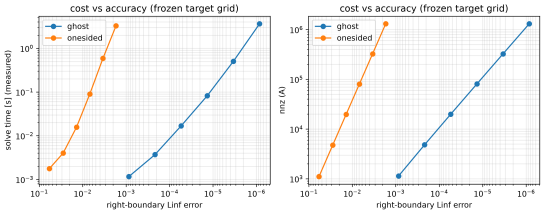
Matrix structure tells the same story



- Five-point sparsity everywhere; the closure only changes one boundary block.
- $\left\|A - A^T\right\| / \|A\|$: ghost closure ≈ 0.02, one-sided ≈ 0.15 — both nonsymmetric, one-sided worse.
- Condition number $\kappa \sim Ch^{-2}$; the one-sided variant has the larger C (M3 predicted this).

Takeaway: structure and conditioning confirm what the errors say — no hidden story.

Experiments: cost vs. accuracy



Frozen target $\|e\|_{\infty,\partial} \le 10^{-4}$:

- Ghost: $N \approx 47$ ($n \approx 2.2 \times 10^3$) — cheap.
- One-sided: $N \approx 8.6 \times 10^3$ ($n \sim 7 \times 10^7$) — infeasible for sparse direct.

At $N = 512$ (same wall time): ghost error 8.5×10^{-7}; one-sided 1.7×10^{-3}.

Takeaway: closure choice is a feasibility choice, not a cosmetic one.

Solvers: direct vs. iterative

Matched algebraic accuracy, Neumann/ghost system at $N = 64$ ($n = 4033$):

Solver	Iterations	Time (s)	Rel. residual	$\|u - u_{\mathrm{splu}}\|_\infty$
Sparse LU (splu)	1 factorisation	0.016	1.5×10^{-15}	0
BiCGSTAB + ILU	39	0.089	8.6×10^{-11}	6.9×10^{-9}
SOR ($\omega = 1.5$)	3721	1.664	1.0×10^{-10}	2.5×10^{-7}
Gauss–Seidel	11192	4.646	1.0×10^{-10}	2.5×10^{-7}
Jacobi	22388	1.979	1.0×10^{-10}	2.5×10^{-7}

Tolerance strategy from Week 7: $\mathrm{tol} \sim Ch^2$ so algebraic error never pollutes discretisation error.

Takeaway: iterative solvers match direct accuracy at lower memory; the closure comparison is independent of the solver choice.

Depth item: pure-Neumann null space

Neumann on the *whole* boundary $\Rightarrow A\mathbf{1} = 0$ exactly (machine zero), smallest left singular vector = constant (2.6×10^{-12}).

Discrete compatibility: $|\langle b, w\rangle|/(\|b\|\,\|w\|) = 5.8 \times 10^{-6}$ — not machine zero: corner rows sample two one-sided data and the trapezoid boundary rule. We report it, not hide it.

Pinned gauge $u(0,0) = 0$ + least-squares constant shift: L^2 order 2.05, L^∞ order 1.84 (corners limit the max-norm rate — bounded, not derived).
Takeaway: the null space is structural, and the recovery is verifiable; an incompatible RHS is caught by the falsification variant.

Conclusions and limitations

Conclusions

1. Two second-order-accurate-looking closures give global orders 2 vs. 1 — the boundary row *is* the method.
2. Ghost elimination buys feasibility: 10^4-fold fewer unknowns for the same boundary accuracy.
3. The evidence chain (row defects $\rightarrow$ refinement tables $\rightarrow$ falsification $\rightarrow$ structure/cost) makes every number auditable; one command (`python run_all.py`) reproduces all of them.

Limitations / open

- Square domain only; corners at non-right angles not treated.
- Pure-Neumann L^∞ order 1.84: attributed to corner quadrature, full derivation open.
- SOR parameter ω not tuned; AMG comparison is future work.

Reproducibility & contributions

One command: `python run_all.py` → all tables, figures, `results.json`.

Team O contributions (git-traceable):

- **Zihan Xu (2360751):** project management, visualisation, report & slides, cost–accuracy study.
- **Beixue Liang (2359957):** hand derivations of both closure rows, Taylor defects, pure-Neumann solvability, falsification design.
- **Yumo Li (2253567):** solver implementation and assembly code, iterative-solver comparison, unit tests.

Thank you — questions welcome.